Teaching the Next Web

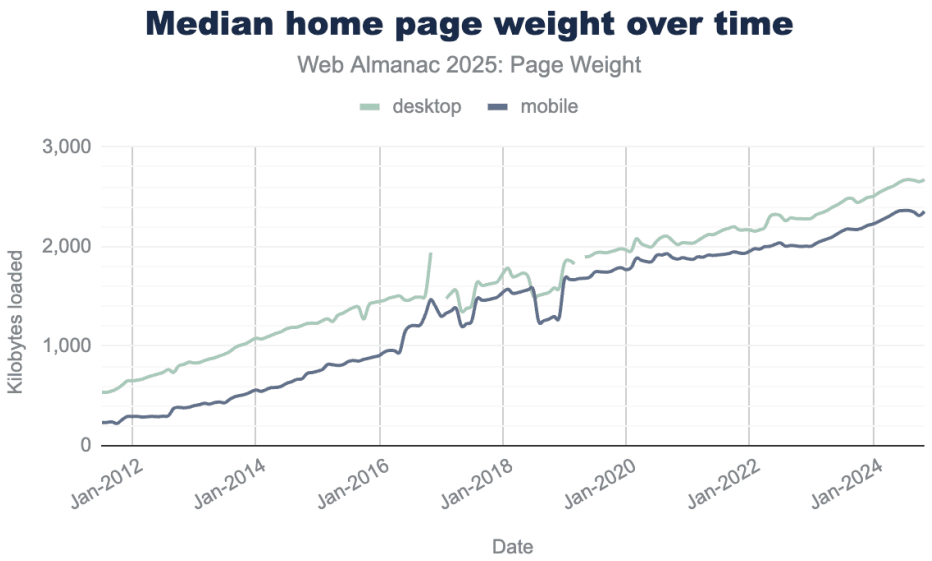
From hybrid rendering to agentic hypermedia

Juho Vepsäläinen

6.5.2026

MOTIVATION

Why this matters now



Source: HTTP Archive Web Almanac 2025, Page Weight, Fig. 14.1

MOTIVATION

Why this matters now

From human browsing to delegated interaction

Search and click

The old interaction model is link-following and interpretation. [6]

Agentic search

Systems now do part of the browsing and synthesis work. [7], [8]

Browser use

Websites are becoming indirect machine interfaces. [9], [10]

Sources: [6]-[10] See references.

Outline

Part 1	Research highlights
Part 2	From research to teaching
Part 3	Future vision and goals
Part 4	Conclusion

Part 1

Research highlights

HIGHLIGHT 1

Hybrid rendering models

Dissertation: Vepsäläinen, J. (2025). Emergence of hybrid rendering models in web application development.

Dissertation background

From static content to web applications.

The platform was designed for static content, then evolved into an application platform.

Choose rendering strategies case by case.

Application size tripled in a decade while mobile constraints remained.

Main claims

Rendering should follow use cases.

The mainstream SPA model should be used more sparingly.

Hybrid rendering narrows dynamic behavior to where it is needed.

Islands constrain dynamic scope. Resumability avoids hydration and enables code-splitting.

Teaching consequence: students need architectural principles more than framework preferences.

HIGHLIGHT 1

Hybrid rendering techniques

The Potential of Serverless Edge-powered Islands for Web Development
Juho Vepsäläinen, Petri Vuorimaa, Arto Hellas • Journal of Web Engineering • 2025

Resumability - a New Primitive for Developing Web Applications
Juho Vepsäläinen, Misko Hevery, Petri Vuorimaa • IEEE Access • 2024

Implications of Edge Computing for Static Site Generation
Juho Vepsäläinen, Arto Hellas, Petri Vuorimaa • WEBIST • 2023

HIGHLIGHT 2

Simplification agenda

Revisiting Hypermedia, The Forgotten Web Application Development Paradigm
Juho Vepsäläinen • TechRxiv preprint • 2026

The Rise of Disappearing Frameworks in Web Development
Juho Vepsäläinen, Arto Hellas, Petri Vuorimaa • ICWE • 2023

The Case for HTML First Web Development
Juho Vepsäläinen • Journal of Web Engineering (accepted) • 2026

Part 2

From research to teaching

CSED

AI and computing education

Educational implications

- **Curriculum and pedagogy need rethinking.**

What changes in practice

Assessment and classroom design

Flipped learning, oral exams, dialogical assessment.

[1] Vepsäläinen, J., & Juntunen, P. (2026). Artificial intelligence for computer science education - hype or help?

CSED

AI and computing education

Educational implications

- **Curriculum and pedagogy need rethinking.**
- **Integrity is not the only issue.**

What changes in practice

Assessment and classroom design

Flipped learning, oral exams, dialogical assessment.

Beyond integrity

Also trust, over-reliance, bias, and unequal access.

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CSED

AI and computing education

Educational implications

- **Curriculum and pedagogy need rethinking.**
- **Integrity is not the only issue.**
- **The goal is responsible technical judgment.**

What changes in practice

Assessment and classroom design

Flipped learning, oral exams, dialogical assessment.

Beyond integrity

Also trust, over-reliance, bias, and unequal access.

Durable skills

Critical reading, writing, debugging, reflection.

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TEACHING

How research informs my teaching

20+
BSc theses since 2022

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MSc theses advised

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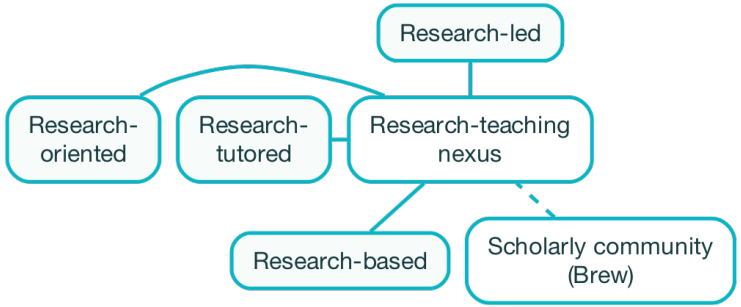
ECTS of pedagogy

Teaching approach

- **Build working models students can adapt.**
  Research helps me teach principles, not only tools.



Research-teaching nexus [3,4]

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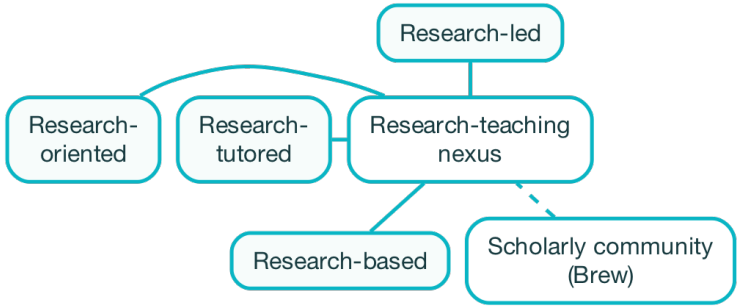
Teaching approach

- **Build working models students can adapt.**
  Research helps me teach principles, not only tools.
- **Teaching and supervision shape my questions.**
  Student work reveals topics that deserve more attention.



Research-teaching nexus [3,4]

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Research with students

- **Student projects can feed a wider research agenda.**

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Pyry Pohjalainen (2025)
AI for web development; AAAI Magazine.

Anoop Bidikar (2024)
Stateful serverless architectures at the edge.

Antti Nousiainen (2024)
WebAssembly in edge computing.

TEACHING

Writing support

A writing process students can use

- **Shared writing process improves supervision.**
  "The Process of Scientific Writing" makes publishing expectations concrete.

- **Students can use it on their own.**
  It supports supervision through a shared writing process.

TEACHING

Public learning communities

Teaching also happens outside courses

- **I organize Future Frontend.**

  Meetups and the conference keep me grounded in current practice.

- **My developer background supports my teaching.**

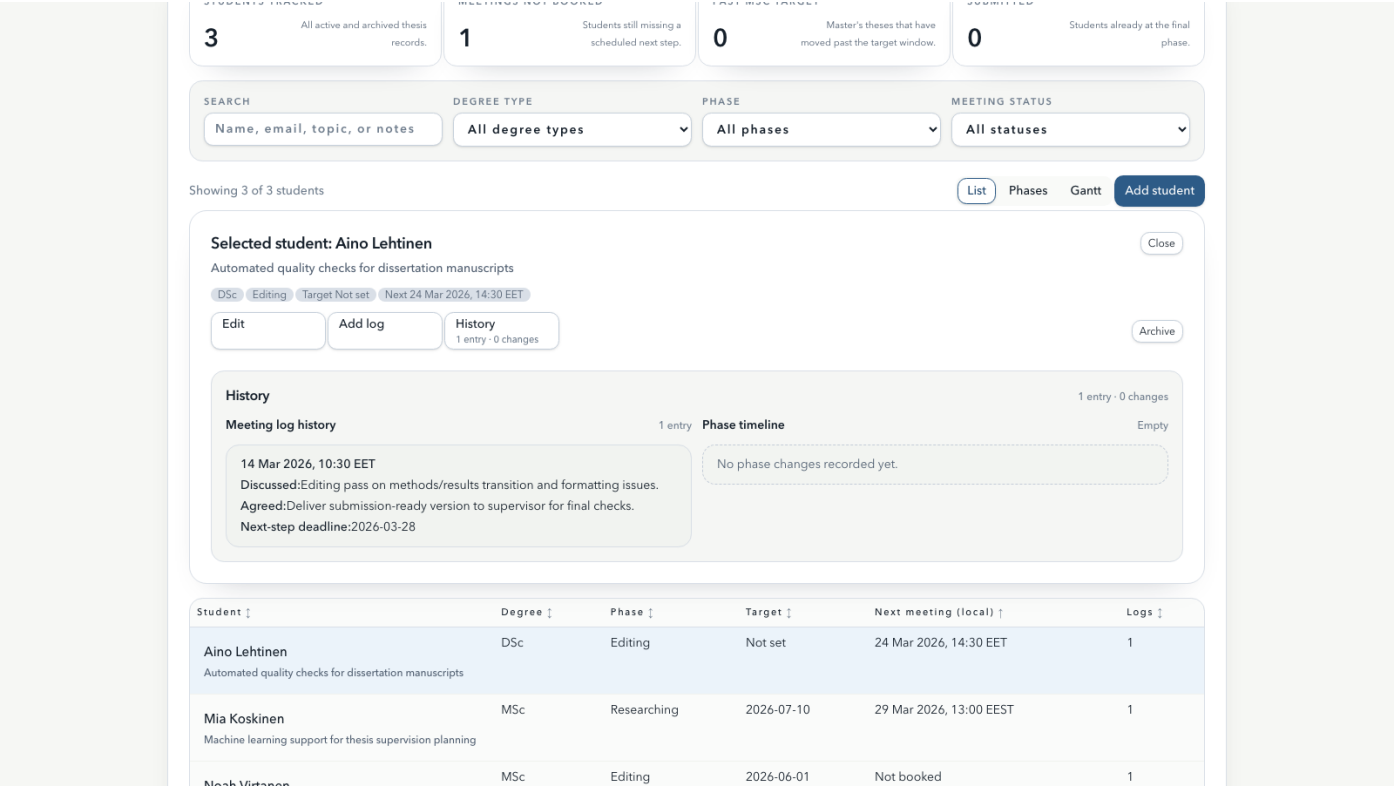
  webpack-merge is one example; earlier work includes webpack and Blender.

Teaching outside the classroom

Future Frontend is one way I do it.

TEACHING

Tools that support my work with students



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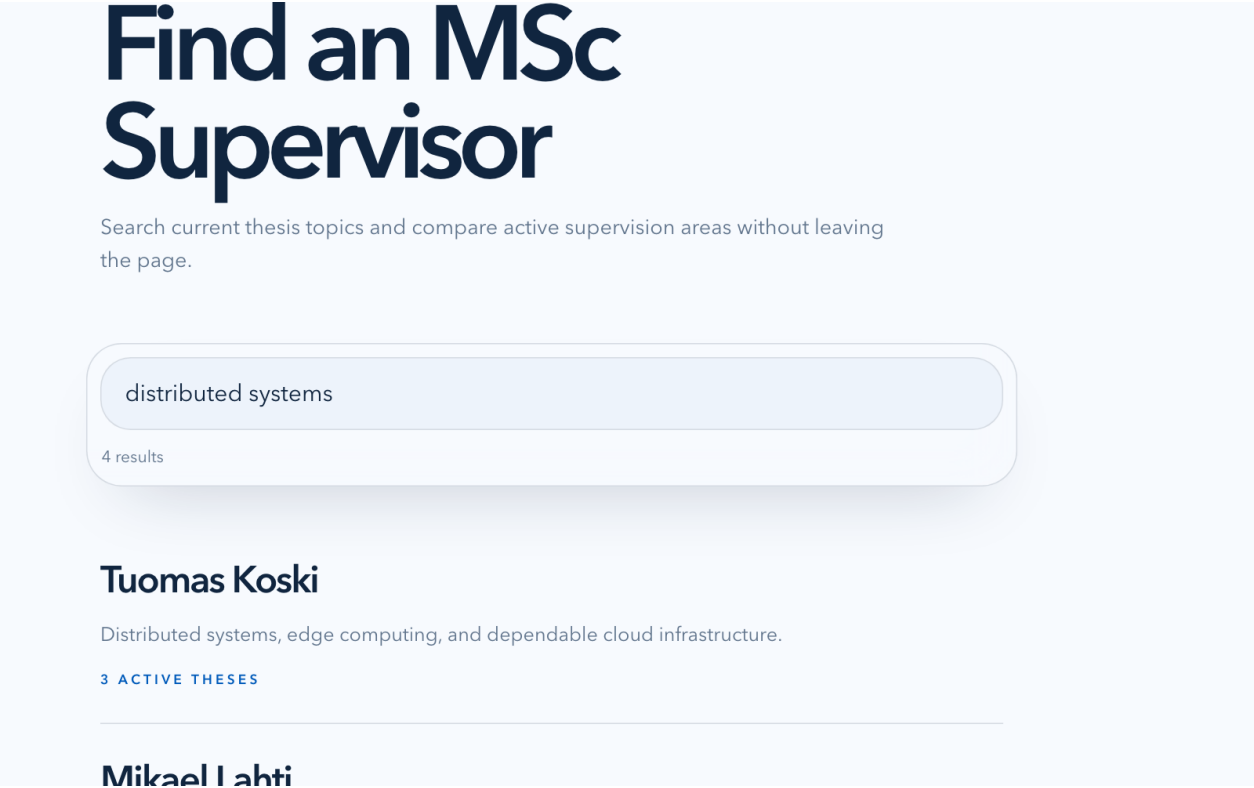


Part 3

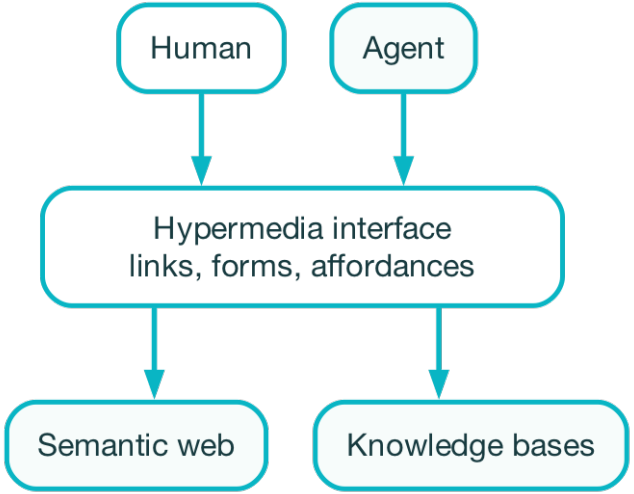
Future vision and goals

VISION

Future vision: agentic hypermedia

Research questions

- **How do we keep the web usable by agents?**
  Recover hypermedia so capabilities stay explicit and navigable.
- **How do the semantic web and knowledge bases matter again?**
  Ground hypermedia in the semantic web and knowledge bases.
- **What should remain stable in web architecture as agents become users?**
  Architecture and judgment matter as tools keep changing.



[2], [5] Fielding; Berners-Lee et al.

CONTRIBUTION

My contribution to Aalto

- **I already connect research, students, and practice at Aalto.**

Agentic/web space
This deepens one technical area within a broad department.

Real systems experience

External ties
I am an active part of the developer community in Finland.

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- **I turn recurring teaching problems into concrete support.**
  I maintain active ties to both academic and industry communities.

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Part 4

Conclusion

CLOSING

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What do students still need to understand as tools change?

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How should we teach architecture in an agentic world?

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What do students still need to understand as tools change?

How should we teach architecture in an agentic world?

How do we help students build judgment rather than overreliance on AI?

Q&A

SOURCES

References

[1] Vepsäläinen, J., & Juntunen, P. (2026). Artificial intelligence for computer science education - hype or help? Preprint.

[2] Fielding, R. T. (2000). Architectural Styles and the Design of Network-based Software Architectures. PhD dissertation, UC Irvine.https://ics.uci.edu/~fielding/pubs/dissertation/

[3] Healey, M. (2005). Linking research and teaching to benefit student learning. Journal of Geography in Higher Education, 29(2), 183-201.https://doi.org/10.1080/03098260500130387

[4] Brew, A. (2012). Teaching and research: new relationships and inquiry-based higher education. Higher Education Research & Development, 31(1), 101-114.https://doi.org/10.1080/07294360.2012.642844

[5] Berners-Lee, T., Hendler, J., & Lassila, O. (2001). The Semantic Web. Scientific American, 284(5), 34-43.https://lassila.org/publications/2001/SciAm.html

[6] Gou, B., Huang, Z., Ning, Y., et al. (2025). Mind2Web 2: Evaluating Agentic Search with Agent-as-a-Judge. NeurIPS Datasets & Benchmarks.https://osu-nlp-group.github.io/Mind2Web-2/

[7] OpenAI. (2025). Introducing deep research.https://openai.com/index/introducing-deep-research/

[8] OpenAI. (2025). Introducing Operator.https://openai.com/index/introducing-operator/

[9] Anthropic. (2024). Introducing computer use, a new Claude 3.5 Sonnet, and Claude 3.5 Haiku.https://www.anthropic.com/news/3-5-models-and-computer-use

[10] Google. (2024). Google introduces Gemini 2.0: A new AI model for the agentic era.https://blog.google/innovation-and-ai/models-and-research/google-deepmind/google-gemini-ai-update-december-2024/